

End2Race: Efficient End-to-End Imitation Learning for Real-Time F1Tenth Racing

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Abstract—F1Tenth is a widely adopted reduced-scale platform for developing and testing autonomous racing algorithms, hosting annual competitions worldwide. With high operating speeds, dynamic environments, and head-to-head interactions, autonomous racing requires algorithms that diverge from those in classical autonomous driving. Training such algorithms is particularly challenging: the need for rapid decision-making at high speeds severely limits model capacity. To address this, we propose End2Race, a novel end-to-end imitation learning algorithm designed for head-to-head autonomous racing. End2Race leverages a Gated Recurrent Unit (GRU) architecture to capture continuous temporal dependencies, enabling both short-term responsiveness and long-term strategic planning. We also adopt a sigmoid-based normalization function that transforms raw LiDAR scans into spatial pressure tokens, facilitating effective model training and convergence. The algorithm is extremely efficient, achieving an inference time of less than 0.5 milliseconds on a consumer-class GPU. Experiments in the F1Tenth simulator demonstrate that End2Race achieves a 94.2% safety rate across 2,400 overtaking scenarios, each with an 8-second time limit, and successfully completes overtakes in 59.2% of cases. This surpasses previous methods and establishes ours as a leading solution for the F1Tenth racing testbed. Code is available at <https://github.com/michigan-traffic-lab/End2Race>.

I. INTRODUCTION

Autonomous racing has emerged as a compelling testbed for advancing autonomous vehicle technologies. It combines perception, planning, and control under high-speed, interactive, and computationally constrained conditions. Enhancing algorithmic capabilities in this domain is therefore relevant to the broader development of autonomous driving technologies. F1Tenth [1], a 1/10-scale vehicular platform, provides an accessible testbed for autonomous racing education and research. Equipped with a 2D LiDAR [2] or RGB-D camera [3] for perception, the vehicle can reach speeds of up to 17.9 m/s, equivalent to approximately 400 mph at full scale [4]. Each year, competitions are held at venues worldwide and attract broad participation from the community.

Existing approaches for autonomous racing can be broadly grouped into map-based, reactive, and learning-based. Of these, map-based methods are the most commonly used in F1Tenth competitions. They rely on a pre-constructed map to follow a globally optimized raceline while adapting to

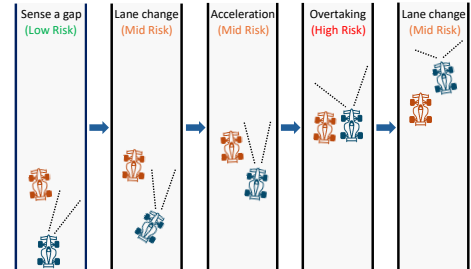


Fig. 1. Simple overtake scenario illustration.

local conditions. In the mapping phase, the vehicle is manually controlled along the track to construct the map using SLAM [5]. Raceline optimization [6] then uses the mapped geometry to derive a global reference path. During racing, particle-filter localization [7] combines odometry and LiDAR measurements to estimate the vehicle pose within the map. Using this estimate, a lattice planner [8] generates candidate trajectories that progress along the raceline while satisfying dynamic and collision constraints. Once a trajectory is selected, its velocity profile provides the speed command, while a path-tracking controller, such as Pure Pursuit [9] or model predictive control (MPC) [10], generates the steering command. This design enables the vehicle to anticipate upcoming curves and consistently track the reference path. Yet map-based methods have two principal limitations. First, they require a preprocessing stage and cannot operate on previously unseen tracks. Second, on an F1Tenth vehicle’s onboard computer [4], localization [11] and planning [12] together incur a systematic delay of approximately 30 milliseconds. With additional sensor latency, the vehicle can travel up to a full vehicle length before updated control commands take effect. This delay is non-critical under most operating conditions but becomes consequential when a high-speed straight segment is directly followed by a sharp curve, leaving the vehicle susceptible to collisions with the track boundaries. Such collisions are widely observed in F1Tenth competitions, including during semifinal and final rounds.

Reactive methods provide a simpler, rule-based alternative by acting directly on current sensor measurements. Their map-free design avoids localization and planning, reducing computational cost while allowing operation on unknown tracks. The Follow-the-Gap algorithm [13], for example, identifies the largest free region and steers toward its midpoint. However, these methods operate instantaneously, as each decision is based on an isolated observation with limited temporal context, leading to planning inconsistencies and control instability [14].

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Learning-based methods have also been widely proposed for F1Tenth. Early work focused primarily on single-vehicle navigation [15], [16], whereas more recent studies have shifted toward head-to-head competition. However, existing approaches either use only static obstacles during training before being tested directly against dynamic opponents [17], rely on a small set of human demonstrations and evaluation scenarios [18], or operate at low racing speeds [19].

In high-speed head-to-head competition, overtaking is a key determinant of the race outcome and presents an intrinsically challenging problem. Figure 1 illustrates a simplified five-stage overtaking process. The ego vehicle first identifies a suitable opening and then executes a sharp lane change to align with it while accelerating rapidly to pass, increasing the risk of losing control. Throughout the pass, the vehicles remain side by side for an extended period, with little clearance between them and from the track boundaries. Finally, the ego vehicle must quickly return to the track center to avoid a boundary collision. Successful overtaking therefore depends on continuous decision-making, precise maneuver execution, rapid adaptation to the opponent’s motion, and minimal sensing-to-control latency. This requires both sufficient model capacity to represent complex conditions and low inference latency, a combination not yet achieved by existing approaches.

To address these limitations, we propose **End2Race**, an end-to-end learning framework for head-to-head autonomous racing that enables systematic training and evaluation of overtaking policies at scale. End2Race takes a 360-degree 2D LiDAR scan and the current vehicle state as inputs and directly produces control actions. To encode temporal dependencies efficiently, we adopt a Gated Recurrent Unit (GRU) network [20] that maintains a hidden state summarizing the observation history. On a physical F1Tenth vehicle’s onboard computer, the network produces control actions with an inference time below 1 millisecond. Training follows a two-stage pipeline in which the policy is first trained through behavioral cloning [21] using demonstrations generated by a map-based expert and then fine-tuned through reinforcement learning using PPO [22]. Across both seen and unseen tracks, End2Race demonstrates strong single-vehicle racing performance as well as competitive overtaking skills.

The main contributions of this work are as follows:

- 1) We introduce End2Race, an end-to-end learning framework for head-to-head autonomous racing that enables systematic training and evaluation of overtaking scenarios at scale.
- 2) We design a computationally efficient recurrent policy network with a sensing-to-control latency below 1 ms, improving safety and supporting real-time operation at high racing speeds.
- 3) Through a two-stage training pipeline combining behavioral cloning with reinforcement learning fine-tuning, End2Race generalizes across unseen tracks and demonstrates strong single-vehicle and head-to-head racing performance.

II. RELATED WORK

A. Imitation Learning

Imitation learning (IL) trains a driving policy from expert demonstrations. Sun et al. [15] generated action labels using a Pure Pursuit controller [9], then evaluated behavioral cloning [21], DAgger [23], HG-DAgger [24], and expert intervention learning [25]. All four policies completed the training track but failed to complete an unseen track. Paluch et al. [26] trained a feedforward policy to imitate nonlinear model predictive control along a precomputed raceline. The learned policy completed repeated laps faster than Pure Pursuit. TinyLidarNet [17] trained a compact convolutional policy from human demonstrations involving only static obstacles and then evaluated it directly against dynamic opponents. Zhang and Loidl [18] trained dense and recurrent policies from human overtaking demonstrations, but their study used limited training data and evaluated the policies using only a small set of scenarios.

B. Model-Free Reinforcement Learning

Model-free reinforcement learning (RL) learns a control policy from observations to maximize cumulative reward. Hamilton et al. [27] compared DDPG [28], SAC [29], and behavioral cloning [21], finding that the RL policies performed better in racing but incurred more collisions. Steiner et al. [19] trained TD3 [30] for head-to-head racing and showed that opponent-inclusive training improved overtaking success against multiple controllers. However, their ego vehicle was limited to 2 meters per second, far below practical racing speeds. High-speed racing constitutes a different regime in which shorter reaction times and stronger dynamic effects demand greater closed-loop stability, real-time responsiveness, and maneuver precision.

C. Model-Based Reinforcement Learning

Model-based RL learns how the vehicle and environment respond to actions, then uses this predictive model to evaluate possible future outcomes. Dreamer [31], for example, learns a recurrent latent dynamics model and trains the policy using trajectories imagined in that latent state. Brunnbauer et al. [16] applied Dreamer to autonomous racing and outperformed model-free agents in task completion and generalization. Meanwhile, performance on unseen tracks varied with the policy inputs.

D. Hybrid Learning Methods

Hybrid methods retain a structured planner or controller within the learned pipeline. Dwivedi et al. [32] incorporated reference trajectories from an external planner into an RL controller, allowing the policy to learn continuous control while retaining high-level planning guidance. Bhargav et al. [33] estimated overtaking success across lateral positions for each track segment and used these estimates to select normal, left, or right planner modes. Trumpp et al. [34] combined residual policy learning with an artificial-potential-field planner for mapless multi-agent racing.

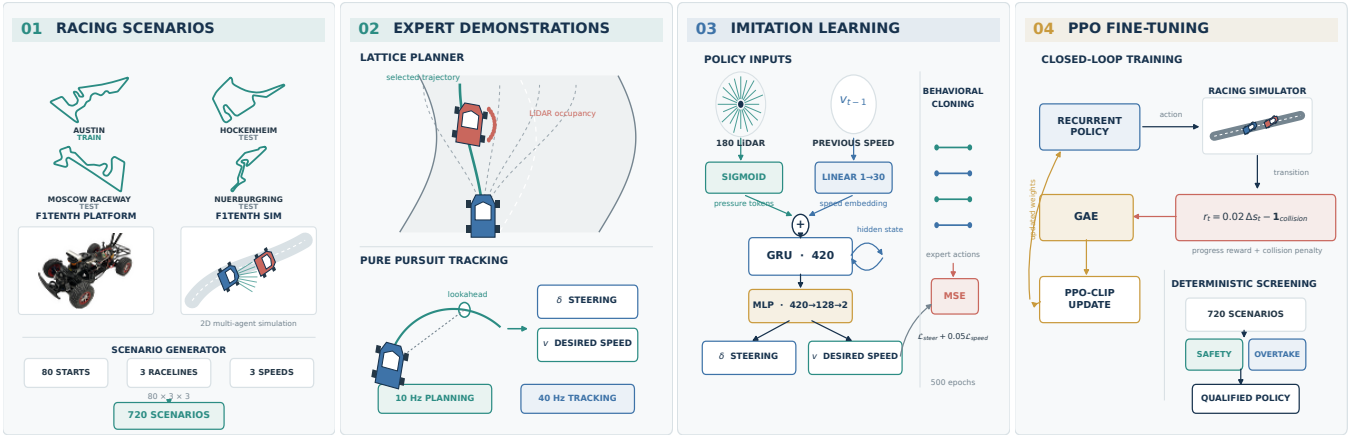


Fig. 2. System overview. The FITENTH vehicle photograph in panel 1 is adapted from the official FITENTH documentation under CC BY-NC-SA 4.0.

III. METHODOLOGY

We develop and evaluate End2Race using the FITENTH Gym simulator [1], which provides high-fidelity vehicle dynamics and accurate 2D LiDAR simulation. Four racing tracks are selected, each reflecting the layout of a real-world Formula 1 circuit: Austin, Hockenheim, MoscowRaceway, and Nuerburgring (Fig. 2(a)). The Austin track is used for training, while the remaining tracks are reserved for evaluation. Within this environment, an automated workflow generates overtaking scenarios at scale across diverse track geometries, raceline configurations, and racing speeds, with each scenario pairing an ego vehicle with a leading opponent. A map-based expert supports scenario execution by generating demonstrations for the ego policy while controlling the opponent (Fig. 2(b)). We first train a Gated Recurrent Unit (GRU) network on these demonstrations through behavioral cloning (Fig. 2(c)), and then fine-tune the resulting policy through reinforcement learning with PPO (Fig. 2(d)). The following sections describe each stage in detail.

A. Overtaking Scenario Generation

Each overtaking scenario pairs an ego vehicle with an opponent initially positioned ahead of it and has a duration of 8 seconds. Vehicle placement is expressed in Frenet coordinates [35], in which the longitudinal direction measures progress around the circuit and the lateral direction specifies position across the track width. We specify each scenario using four parameters, $(s_0, d_0, r_{\text{opp}}, \alpha_{\text{opp}})$, as detailed below. **Ego Starting Position** (s_0). For each track, we define N_s equally spaced ego starting positions and fix $N_s = 80$ across all tracks. For example, on the 420 m Austin track, adjacent starting positions are separated by approximately 5.2 m in the longitudinal direction.

Initial Distance (d_0). The initial distance d_0 specifies how far ahead of the ego vehicle the opponent is positioned at the start of a scenario. A short distance provides insufficient starting space for the ego vehicle to maneuver. Conversely, a long distance requires the ego vehicle to spend more time catching the opponent, leaving less time for interaction. We therefore set $d_0 = 3$ m to balance these considerations.

Opponent Raceline (r_{opp}). Beyond their longitudinal placement, the relative lateral positions of the ego vehicle and opponent are critical to the overtaking geometry. We specify these positions through reference racelines, which define the paths followed by the vehicles as they progress along the track. To represent this variation, the center reference is placed at 50% of the local usable track width, while the left and right references are placed at 35% and 65%, respectively. The usable width excludes a 0.3 m safety margin along each track boundary. For example, the Austin track has an average usable width of approximately 1.64 m, producing an average lateral separation of approximately 0.25 m between the center reference and each side reference. A raceline is generated for each reference following the method in [36], and r_{opp} specifies which raceline is assigned to the opponent. The ego uses the center raceline as its nominal reference but may temporarily deviate from it to overtake or follow a dynamically advantageous path through a curve. The opponent is assigned equally across all three racelines. An opponent following the center raceline leaves limited clearance on either side, whereas either side raceline generally provides a wider passing space on the opposite side.

Opponent Speed Scale (α_{opp}). Let $v_{r_{\text{opp}}}(s)$ denote the curvature-dependent desired-speed profile of the selected opponent raceline, capped by the vehicle's maximum speed of 7.5 m/s. The scaled opponent reference speed is

$$v_{\text{opp}}^{\text{ref}}(s) = \alpha_{\text{opp}} v_{r_{\text{opp}}}(s). \quad (1)$$

We set α_{opp} to 0.4, 0.6, or 0.8, while the ego retains a speed scale of 1.0. Empirical testing showed that scales below 0.4 produce opponents that are unrealistically slow for competition, whereas scales above 0.8 leave few opportunities for the ego to catch the opponent within 8 seconds. The scale modifies the opponent's desired speed rather than its realized speed, which also depends on local curvature, acceleration limits, and closed-loop tracking.

With d_0 fixed, combining $N_s = 80$ values of s_0 , three values of r_{opp} , and three values of α_{opp} yields 720 scenarios per track.

B. Expert Demonstration

In each scenario, both the ego and the leading vehicle are controlled using a lattice planner and Pure Pursuit controller, following the approach described in [37]. The lattice planner generates a discrete set of candidate trajectories and selects the optimal one according to a composite reward function:

$$\mathcal{R} = \lambda_v \ln(v_{\text{ego}}) - \lambda_p d_r - \lambda_d \phi(d_l) - \lambda_\kappa \kappa v_{\text{ego}} \quad (2)$$

where

- v_{ego} : projected speed of the ego vehicle
- d_r : lateral deviation from the reference path r
- d_l : minimum distance to the leading vehicle
- $\phi(d_l)$: non-linear cost on d_l
- κ : road curvature on the reference trajectory
- $\lambda_v, \lambda_p, \lambda_d, \lambda_\kappa$: weighting coefficients

The ego vehicle's trajectory reward is tuned to emphasize higher speeds and opportunistic overtaking, while the leading vehicle is set to follow its path smoothly. In line with prior work [38], the leading vehicle does not engage in adversarial maneuvers such as active blocking or abrupt acceleration.

The Pure Pursuit controller computes the steering angle δ based on a lookahead point:

$$\delta = \arctan\left(\frac{2L \sin \alpha}{\ell}\right) \quad (3)$$

where L is the wheelbase, α is the angle between the vehicle's heading and the lookahead point, and ℓ is the lookahead distance. This ensures reliable path tracking by continuously adjusting steering to minimize deviation from the reference path.

C. Training Dataset

Using the scenario generation process described above, we generate 600 overtaking scenarios per track, each with an 8-second time limit. The Austin track is used to construct the training dataset, while the remaining tracks are reserved for evaluation (Fig. 2). For each training scenario, expert actions for the ego vehicle are recorded and downsampled to 10Hz (from an original simulation frequency of 100 Hz), while the policy of the opponent is not used.

Notably, even a rule-based expert such as the lattice planner cannot guarantee safety under all conditions, reflecting the inherent challenges of real-world autonomous racing. These incidents can stem from minor control imperfections that cause vehicles to drift or from overtaking maneuvers encountering unexpectedly narrow corridors, leaving insufficient space for both vehicles to pass safely.

Furthermore, by design, the lattice planner considers only a predefined set of look-ahead points for trajectory planning and obstacle avoidance, without accounting for vehicles approaching from behind. As a result, it might occasionally prompt premature returns to the center lane immediately after overtaking, which can lead to rear-end sideswipe collisions, especially when the speed difference between the ego vehicle and the opponent is low. This illustrates the inherent difficulties in developing autonomous racing policies that balance

safety and efficiency, even when leveraging sophisticated rule-based approaches.

Out of the 600 generated scenarios, 194 featured safe car-following, 378 resulted in successful overtakes, and 28 ended in collisions. Unsafe demonstrations that result in collisions are excluded from the training set, resulting in a total of 45,760 training samples. However, these scenarios are still included during evaluation to assess model performance.

D. Model Architecture

We adopt a Gated Recurrent Unit (GRU) architecture that maps raw observations to driving commands. At each timestep t , the network receives a 360-degree LiDAR scan $\mathbf{z}_t \in \mathbb{R}^{360}$ at 1-degree resolution and the ego speed $v_t \in \mathbb{R}$. The update is defined as:

$$\mathbf{h}_t = \text{GRU}_\theta(\sigma(\mathbf{z}_t) \parallel \psi(v_t), \mathbf{h}_{t-1}) \quad (4)$$

Here, $\mathbf{h}_t$ denotes the hidden state of the model, and its dimensionality is set to four times the input size to capture the complexities of the racing task.

To better represent the sensor input, the raw LiDAR scan $\mathbf{z}_t$ is normalized elementwise using a sigmoid function $\sigma(\cdot)$, defined as

$$\sigma(x) = \left(-\frac{1}{1 + e^{-kx}} + 1\right) \cdot 2 \quad (5)$$

where x denotes a single LiDAR scan beam of $\mathbf{z}_t$; $k > 0$ determines the effective sensing range.

This normalization maps each raw scan value, $x \in [0, \infty)$, to a spatial pressure token within the range $[0, 1]$. The S-shaped curve of the sigmoid ensures that large scan distances x are mapped to values that asymptotically approach zero, meaning distant obstacles have minimal influence on the network. As obstacles become closer, the token value increases rapidly and nonlinearly, making the system much more sensitive to nearby objects. At the same time, the maximum value is bounded at 1 as x approaches zero, ensuring stable normalization outputs. In this way, the LiDAR inputs are represented in a nonlinear, continuous, and differentiable manner that naturally aligns with human cognition. This facilitates more efficient training and enables the model to converge more effectively.

The ego speed v_t is transformed using a learnable projection $\psi(v_t)$, which encodes the scalar input into a higher-dimensional embedding. It is then concatenated with the normalized spatial pressure tokens to form the observation at each timestep, serving as the GRU input. The recurrent architecture also propagates the previous hidden state, $\mathbf{h}_{t-1}$, to the next timestep, incorporating the latest observations to produce the updated hidden state, $\mathbf{h}_t$. This design enables the model to capture long-term temporal dependencies while maintaining a compact summary of past observations without excessive computation, which is critical for modeling high-speed, dynamic interactions in autonomous racing scenarios.

To predict the driving actions, we apply a decoder head implemented as a two-layer multilayer perceptron (MLP):

$$\begin{aligned} \mathbf{h}'_t &= \text{ReLU}(W_1 \mathbf{h}_t + \mathbf{b}_1), \\ \mathbf{a}_t &= W_2 \mathbf{h}'_t + \mathbf{b}_2 \end{aligned} \quad (6)$$

where $\mathbf{a}_t = [v, \delta]$ denotes the ego speed and steering angle.

E. Training Setup

The model is trained for 500 epochs using the Adam optimizer, initialized with a learning rate of 0.001. To facilitate convergence, a learning rate scheduler monitors the training loss and automatically halves the rate whenever progress stalls. Training is conducted with mini-batches of size 16 and *completes in 10 minutes on a single NVIDIA 4080 GPU*, converging to a near-zero final loss.

To preserve temporal dependencies, each scenario input sequence is considered in its entirety, producing predictions at each timestep and aggregating losses across the full temporal horizon. The loss function is defined as a weighted sum of mean squared errors (MSE) for both the ego vehicle's speed and steering angle, balancing their relative scales and jointly guiding the model toward accurate control.

$$\mathcal{L} = 0.05 \mathcal{L}_{\text{speed}} + \mathcal{L}_{\text{steer}} \quad (7)$$

To prevent shortcut learning [39], where the model simply copies the current ego speed as output, we employ a masking technique that randomly replaces the ego speed embedding with a learnable mask.

$$\psi(v_t) = \begin{cases} \psi(v_t), & \text{with probability } 1 - p \\ \mathbf{e}_{\text{mask}}, & \text{with probability } p \end{cases} \quad (8)$$

During training, we set the masking probability to $p = 0.1$. This strategy encourages the model to extract meaningful patterns from the LiDAR input while still leveraging previous vehicle dynamics for smoother control.

IV. EXPERIMENTS

This section reports experimental results on the training track as well as three unseen test tracks. We begin by presenting the model's performance in the single-agent setting, showing detailed driving metrics. Next, we evaluate its behavior across 2,400 overtaking scenarios spanning all four tracks, analyzing outcomes under diverse conditions. Two comparative examples are shown in Fig. 3 and 4, illustrating a successful overtaking maneuver and safe car-following, respectively. Finally, we conduct stress tests with varying levels of sensor noise, which is essential for ensuring reliable operation in real-world autonomous racing applications.

A. Single-Agent Navigation

For each track, the ego vehicle is evaluated by performing ten consecutive laps, which aligns with standard practice in F1Tenth competitions. Table I summarizes the driving metrics. Across all tracks, the model successfully completes ten laps without collision, even on challenging sections featuring sharp turns and continuous sweeping curves. The mean speed remains high at about 7.34 meters per second,

which corresponds to 165 miles per hour in full-scale racing. The low speed variance indicates consistent driving, with only moderate deceleration at sharper corners, while the low lap time variance further supports this claim. These results alone demonstrate the effectiveness of our model, achieving a level of performance not previously attained by imitation learning methods in the F1Tenth domain.

B. Head-to-Head Racing

Across all 2,400 overtaking scenarios conducted on four challenging tracks, each lasting 8 seconds, the model consistently achieves a high safety rate, exceeding 93.5% on every track and averaging 94.2%. The overtaking success rate is similarly strong, with an average of 59.2% across all tracks. In cases where overtaking cannot be performed safely, the model reliably switches to safe car-following behavior. Moreover, the model demonstrates consistent performance across all tracks, exhibiting negligible difference between the training and evaluation tracks, indicating that it acquires robust racing strategies that generalize beyond the training environment. This result is comparable to the lattice planner expert in terms of both safety and overtaking rate, representing the first imitation learning approach in the F1Tenth literature to reach this level of competency. In addition, our model does not require pre-mapping or fine-tuning, as is necessary for the rule-based expert, and can be deployed directly on any unseen F1Tenth race track.

Relative to prior work [38], which reported an overtaking safety rate of 78.8%, our approach reduces collisions by a substantial margin of 72.6%. More importantly, while their evaluation was conducted on tracks with simple geometries, we deliberately selected tracks with sharp corners and narrow corridors to create more challenging overtaking scenarios.

Note that due to the time constraint in each scenario, overtaking opportunities are inherently limited. Extending the duration (not adopted here to avoid unnecessarily longer training and testing) would substantially increase the overtaking rate of the model. In real F1Tenth competitions, overtakes are rare, and even a single successful maneuver across all 10 laps can provide a decisive advantage over an opponent.

C. Scenario Illustration

We illustrate an overtaking and a car-following scenario with continuous time frames (Fig. 3 and 4). The two scenarios feature nearly identical initial conditions, differing only in the leading vehicle's velocity, highlighting the contrastive behavior. In each case, the vehicles are navigating a continuous sweeping curve that consists of a left turn immediately followed by a right turn. The ego vehicle is shown in blue and the leading vehicle in red, with their trajectories depicted in the corresponding color.

In this setup, the leading vehicle follows the centerline of the track, leaving minimal gaps on either side, which makes overtaking especially challenging. Initially, the ego vehicle follows the leading vehicle (Fig. 3.1). As they approach the upcoming right turn, a brief opening appears, and the ego vehicle seizes the opportunity by steering sharply to the right

TABLE I
SINGLE-VEHICLE PERFORMANCE

	Mean Speed (m/s)	Speed Variance (m/s ²)	Mean LapTime (s)	LapTime Variance (s ²)	Laps Completed
Austin (Train)	6.93	0.78	60.44	1.19	10
Hockenheim	7.40	0.50	48.81	0.34	10
MoscowRaceway	7.21	0.54	44.71	0.11	10
Nuerburgring	7.80	0.39	57.56	0.21	10
Summary	7.34	0.55	52.88	0.46	10

TABLE II
HEAD-TO-HEAD RACING PERFORMANCE

	Car Following	Overtaking	Collision	Overtake Rate (%)	Safety Rate (%)
Austin (Train)	234	341	25	56.8	95.8
Hockenheim	233	330	37	55.0	93.8
MoscowRaceway	194	367	39	61.2	93.5
Nuerburgring	178	384	38	64.0	93.7
Summary	210	355	35	59.2	94.2

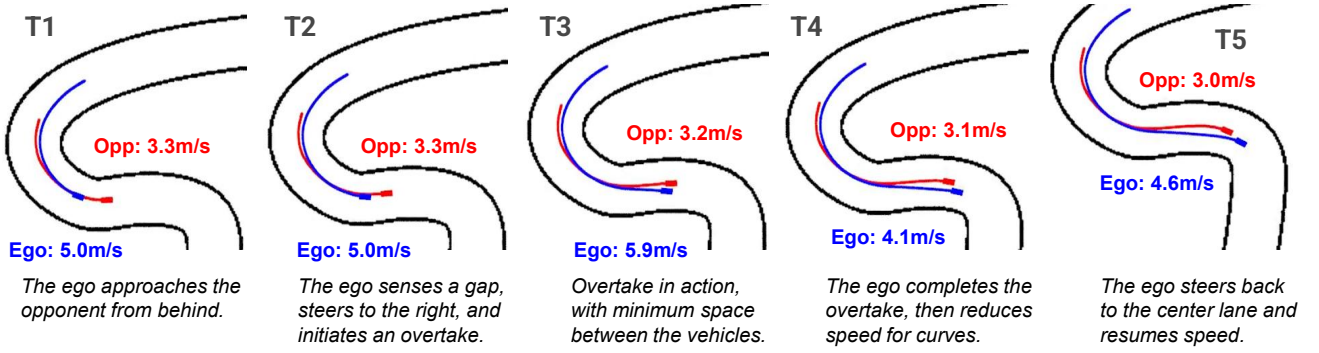


Fig. 3. Overtaking example.

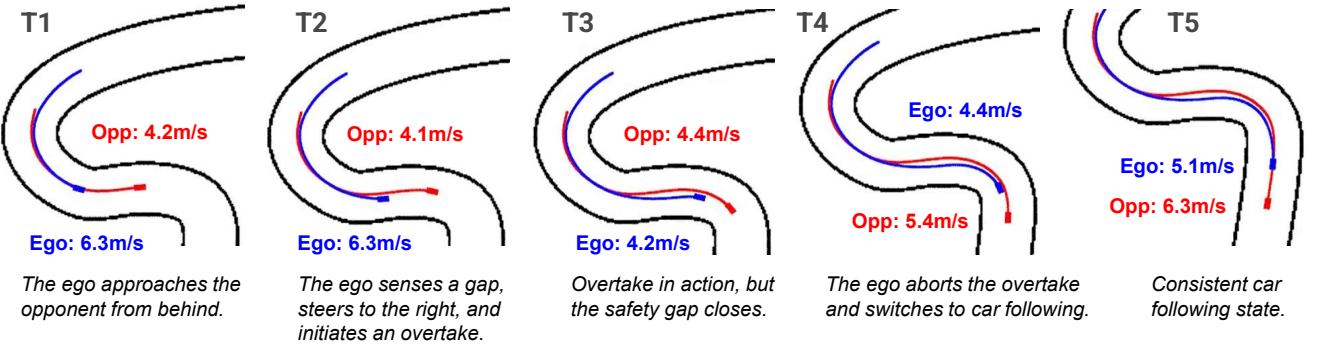


Fig. 4. Car-following example.

and accelerating (Fig. 3.2 and 3.3). Upon entering the turn, the ego vehicle utilizes the inner lane and the shorter path to quickly overtake the leading vehicle. Immediately after passing, it reduces speed to avoid excessive velocity through the curve, which could result in loss of control (Fig. 3.4). After that, the ego vehicle resumes its pace (Fig. 3.5).

In the car-following scenario, the leading vehicle travels at a higher speed and maintains a larger distance ahead of the ego vehicle (Fig. 4.1). As in the overtaking example, the ego

vehicle initially attempts to overtake by steering to the right (Fig. 4.2). However, midway through the maneuver, the gap closes and overtaking becomes infeasible. Instead of forcing an unsafe pass, the ego vehicle aborts the attempt and steers back behind the leading vehicle to resume following (Fig. 4.3 and 4). After both vehicles complete the turn, they accelerate back to cruising speed (Fig. 4.5).

TABLE III
SINGLE-VEHICLE PERFORMANCE (HOCKENHEIM) WITH SENSOR NOISE

	Mean Speed (m/s)	Speed Variance (m/s ²)	Mean LapTime (s)	LapTime Variance (s ²)	Laps Completed
10% Noise	5.48	0.16	65.19	0.21	10
20% Noise	4.41	0.18	80.73	0.28	10
30% Noise	3.85	0.21	92.20	0.47	10
40% Noise	2.76	0.10	-	-	1.4
50% Noise	1.97	0.06	-	-	0.6

TABLE IV
HEAD-TO-HEAD RACING PERFORMANCE (HOCKENHEIM) WITH SENSOR NOISE

	Car Following	Overtaking	Collision	Overtake Rate (%)	Safety Rate (%)
10% Noise	421	155	24	25.8	96.0
20% Noise	535	45	20	7.5	96.7
30% Noise	583	2	15	0.4	97.5
40% Noise	579	1	20	0.2	96.7
50% Noise	553	0	47	0.0	92.2

D. Evaluation with Sensor Noise

In real-world autonomous racing, sensor noise remains a persistent and significant challenge. A common issue is the intermittent return of zeros from some LiDAR beams, caused by hardware malfunctions, environmental interference, or communication errors [40].

These anomalies can significantly degrade system performance. For example, in reactive methods such as Follow-the-Gap, the absence of even a single beam across a wide gap may entirely obstruct the passage, resulting in suboptimal path selection. In map-based methods like the lattice planner, missing measurements can cause localization errors and often lead the vehicle to collide with track boundaries. To the best of our knowledge, the impact of sensor noise has not been investigated in existing F1Tenth learning algorithms.

We assess the robustness of our model by randomly setting a proportion of LiDAR beams to zero at each timestep. Performance metrics are summarized in Tables III and IV, using the Hockenheim track as a representative example.

1) *Single-Vehicle Setting*: Our model demonstrates strong robustness to sensor noise, successfully completing ten laps with up to 30% noise. That is, 108 out of 360 LiDAR beams are set to zero at every timestep, imposing conditions far more demanding than typical real-world sensor failures. Even under 40% noise, the model still completes more than one lap, and under 50% noise, more than half a lap. As noise increases, the model adopts a conservative driving strategy, gradually reducing its speed while maintaining a stable speed variance. This is because our LiDAR spatial pressure token design exerts maximum pressure in the directions affected by sensor noise, prompting the ego vehicle to slow down rather than losing control, thereby preserving smooth and consistent driving behavior even under severe occlusion.

2) *Head-to-Head Racing*: In head-to-head racing scenarios, our model maintains a high safety rate. As the noise level increases, the model performs fewer overtaking maneuvers

and more car-following cases. This indicates that even under heavy sensor noise, the model does not abruptly collide with the opponent. Instead, it slows down and remains cautious until conditions improve.

E. Limitations

Upon reviewing the overtaking cases, we observed that most collisions occur when the ego vehicle returns to the center lane too quickly after completing an overtake, causing its rear to sideswipe the opponent. This behavior does not reflect an inherent limitation of our model, but can be attributed to the imperfect demonstrations of the lattice planner expert, as discussed in previous sections. While such unsafe cases are filtered during training, our model never explicitly learns how to avoid them, and is reflected in evaluation. In future work, a more comprehensive expert design that accounts for post-overtake interactions could enable the model to learn safer strategies and further reduce collision rates.

F. Ablation Studies

This section presents ablation studies on key architectural design choices. We evaluate two GRU variants with hidden sizes of $2\times$ (small) and $8\times$ (large) the input dimensionality, as well as a variant that omits the ego-speed input (LiDAR-only). Evaluations are conducted on the Hockenheim track as a representative example.

The small variant achieves an overtaking rate of 55.0%, similar to the baseline, but its safety rate drops to 86.2%, indicating insufficient model capacity to reliably learn car-following and overtaking behavior. The large variant maintains a high safety rate of 92.7% but is overly conservative, with an overtaking rate of only 48.2%. This suggests possible overfitting to training and limited generalization of overtaking skills. The LiDAR-only variant yields higher overtaking rates (60.3%) but at the cost of safety (90.2%). Without

explicit ego-speed input, the network issues impulsive commands that deviate from the current state, leading to riskier maneuvers. Since safety is paramount in autonomous racing, we retain ego-speed input to ensure a more balanced policy.

V. CONCLUSION

In this work, we introduce End2Race, an innovative end-to-end imitation learning algorithm for F1Tenth racing. Our approach achieves high efficiency in single-vehicle settings, adaptive decision-making during overtaking, and strong robustness to sensor noise. Collectively, these results establish it as a competitive autonomous racing solution. Future work will focus on deployment to a physical vehicle and participation in F1Tenth competitions.

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